

```
# Import libraries

import pandas as pd

# Load dataset

df = pd.read_csv("/content/Employee_Salary.csv")

print("Dataset Loaded Successfully!\n")

# Basic info

print("Dataset Info:")
print(df.info())

print("\nShape of Dataset:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

# Missing values

print("\nMissing Values:")
print(df.isnull().sum())

# Fill missing values

df['Experience'] = df['Experience'].fillna(df['Experience'].mode()[0])
df['Gender'] = df['Gender'].fillna(df['Gender'].mode()[0])

print("\nAfter Handling Missing Values:")
print(df.isnull().sum())
```

```
# Statistical summary
```

```
print("\nStatistical Summary:")
```

```
print(df.describe())
```

```
#  REQUIRED PART: Grouped Summary Statistics
```

```
print("\nSummary Statistics of Salary grouped by Gender:")
```

```
grouped_stats = df.groupby('Gender')['Salary'].agg(['mean', 'median', 'min', 'max', 'std'])
```

```
print(grouped_stats)
```

```
#  REQUIRED PART: Create list
```

```
print("\nList of Salary values for each Gender:")
```

```
salary_list = df.groupby('Gender')['Salary'].apply(list)
```

```
print(salary_list)
```

part 2

```
# Import libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Load dataset (from Colab upload)
```

```
df = pd.read_csv("/content/iris.csv")
```

```
print("Dataset Loaded Successfully!\n")
```

```
# Check column names (IMPORTANT)
```

```
print("Columns in dataset:")
```

```
print(df.columns)
```

```

# Basic information
print("\nDataset Info:")
print(df.info())

print("\nShape:", df.shape)
print("Size:", df.size)

# Display data
print("\nFirst 5 rows:")
print(df.head())

print("\nLast 5 rows:")
print(df.tail())

# Check null values
print("\nNull Values:")
print(df.isnull().sum())

# Statistical summary
print("\nOverall Statistical Summary:")
print(df.describe())

# =====

# REQUIRED PART

# =====

print("\n==== Species-wise Statistics =====")

# Group by species and calculate statistics
species_stats = df.groupby('species').agg({

```

```
'sepal_length': ['mean', 'std', 'min', 'max'],  
'sepal_width': ['mean', 'std', 'min', 'max'],  
'petal_length': ['mean', 'std', 'min', 'max'],  
'petal_width': ['mean', 'std', 'min', 'max']  
})
```

```
print(species_stats)
```

```
# Percentiles
```

```
print("\n===== Percentiles (25%, 50%, 75%) =====")  
percentiles = df.groupby('species').quantile([0.25, 0.5, 0.75])  
print(percentiles)
```